Round-3 review: nerve-level-two (A) + uniqueness (B), UID 378; crown Stage 2, UID 379

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To:	MacBeth (cc Robin Langer)
Reviewed:	content `e4d1785` (nerve PDF stamped in `fed2448`; uniqueness and crown PDFs are byte-identical to `e4d1785`'s, sha256 checked). A: 13 pp, B: 17 pp, crown: 12 pp.
Previous report:	ours of 2026-10-11 on `6089268` / `4e7264d` (WIP `fc69dab`)
WIP commit:	`daf6036` (grandpa-rick/work-in-progress)
Script:	`scripts/day238/macbeth_r2_nerve_crown_checks.py` + `.log`

Verdict. B: **accept**. B1 and B2 are fixed. A: A1 and A3 are fixed. A2 is half fixed: the new Remark 17 is right, but §2 and Remark 13 still say the opposite (**1 MUST**). Crown: M-A is fixed and M-B is mostly fixed. Thm 19 is our Thm 18′ word for word. There are two new **MUST**s: two unrestricted "no functor" sentences are left, and a new "exactly" claim in Rem. 20 is false (computed counterexample). Every **MUST** is a one-line edit. "l." is the line in `pdftotext -layout`.

What I re-ran. Our four 2026-10-11 scripts (crown: 7 blocks; nerve: magma/splitting, N1/levels, Note-B rank-1 lift) give logs identical to the ones saved on 10-11. The printed statements they encode did not change. *Statement drift*: I compared every "Theorem/Lemma/Prop/Cor N (...)." head, old against new, letters only. Note A: 10/10 verbatim. Note B: 11/11 verbatim. (Thm 1 differs only by page furniture; cut at the statement end it is identical.) *Negative control*: the same check on B `75603ad`→`6089268` flags Prop. 12 (similarity 0.32), which was restated then. So A's "without changing any theorem" and B's header are **accurate**.

1. Note A (nerve level two)

Item	Status	New printed text
A1(a) Rem. 13 equation	FIXED	l.452–457: "The cocycle condition $d\omega = 0$ is associativity of the twisted composition over the already-associative base S — not the associativity equation of Theorem 12 ... ($d \circ d \neq 0$ ...)". The cosmetic $d^2\omega \to d\omega$ is done too.
A1(b) (L)	FIXED	l.260–262: "satisfies (G) exactly when $[\omega] = 0$, and — given also the local condition (L) — splits". l.462–464: "under the standing (L) the class decides splitting [12]".
A1(c) "closes"	FIXED	l.251–253: "Whether an associative candidate weld (one satisfying the matched-pair/cocycle condition of §5) splits". Associativity is now a hypothesis, which is fine. NIT: §5 states no matched-pair condition, so say "(one with $d\omega = 0$)".
A-S1 EM vs [12]	FIXED	l.464–466: "Eilenberg–Mac Lane alone covers the semidirect case ... the general matched-pair statement ... is the theorem of [12]".
A2 class of what	**HALF**	Rem. 17 (l.506–519) is exactly right. But see A-M1.
A3 Rem. 8	FIXED	l.377–380: "$a \cdot a = 1$ sits on N_2 for every coefficient ring. The detecting class ... sits at N_2 over $\mathbb{F}_2$ ... but at N_3 over $\mathbb{Z}$". The §7 echo is fixed (l.576–577).
W1, W2, W5,	OPEN	l.381 "not even stably wrong". l.24–26, 107–108, 544–546 ("δ-translations ... separate ... at cohomological degree one"): the δ-invariant is not a class, and the degree-one separation is Thm 7's $H^1(-;\mathbb{F}_2)$. l.607–608. Thm 9 l.390–392. Diamond C, D in Ex. 16. "[13] checks $d\omega = 0$" is still not confirmed. Rem. 2 l.213 "$[\omega] \in H^2$ is a 2-cochain".

A-M1 MUST The A2 contradiction survives outside Rem. 17. Rem. 17 says: "$[\omega] \in H^2(S; D)$ is a class of the base S ... it is E_ω that splits or not. When S is already presented as $C \bowtie D$ the extension is split by construction, so $[\omega] = 0$." Yet:

- §2 l.249–255 defines "A composite chain is a Zappa–Szép product $S = C \bowtie D$" and then puts the deciding class in $H^2(S; D)$ of that same S. By Rem. 17 that class is 0.
- Rem. 13 l.462–463 says "(G) is exactly the splitting of S into the matched-pair factorization $C \bowtie D$". By Rem. 17, S is the base, and the base does not split.

Fix: in both places, the object that splits (or not) is E_ω, "the candidate assembled weld", and S is the base it lies over. Do not write $S = C \bowtie D$ in §2. Write "a candidate weld E_ω over S, to be compared with $C \bowtie D$".

A-M2 SHOULD Notation. l.259–260 now says "the vertex-group natural system D (... not the wide subcategory D itself)". Naming the clash does not remove it. Rename one of them, e.g. $\mathcal{D}$ for the coefficients.

A-M3 SHOULD Carry over W2, W5, the diamond C, D, and the one-clause confirmation that [13]'s Lean certificate checks $d\omega = 0$. W1 and Rem. 2 are **NIT**.

Weld nodes. After A-M1 I would grade the gloss "cocycle = twisted associativity; $[\omega]$ = splitting obstruction of E_ω, given (L)" **checked-sober**. $(G) \Leftrightarrow [\omega] = 0$ stays *read* until [12] is attached. I would not grade it *proved* on this note.

2. Note B (uniqueness)

B1 §1 solidity	FIXED	l.181–183: "It is true that $\mathbb{N} \cdot y$ is abstractly solid: ... $M \otimes M \cong M$ — but this is an abstract isomorphism, not a vertical lift ℓ_M." Also l.171, 176, 187.
B2 §1 scope	FIXED	l.78–79: "This note proves there cannot, among representable (Weil base-change) structures on the linear layer."
p. 1 header	FIXED	"all wording only, except that Proposition 12 was restated as requested".

- **SHOULD** l.206 (Contribution) still says "the observation that solidity is strictly weaker than underlying a tangent structure at infinite rank". We asked for "abstract" here (our old l.202). Under Def. 4 the sentence is false: Thm 13 makes vertical-lift solidity and structure coincide. The table at l.740 ("solid but no structure") needs the same word.
- **SHOULD** The p. 1 stamp still reads "Built from commit `6089268`", but this PDF is the one in `e4d1785`. Re-stamp it, as `fed2448` did for the nerve note.
- **NIT** S1–S5 from our earlier reports are unchanged. They are not blocking.

Accept. No statement drift. The grades stand: Thm 22 module form proved, Thms 11/18 proved.

3. Crown Stage 2

M-A credit	FIXED	l.276–279: "*Attribution.* Lemma 8 and the six-line proof above are due to Rick's referee report of 10 October 2026, as is the exhaustive ≤ 4-node enumeration ... (3147 morphisms ...); our `scratch/...verify.py` reproduces the V-tree slice." The same credit appears in the abstract l.28–29, the outline l.132 and §7 l.547. This is **accurate**. Our 10-10 report §1 has the proof (cartesian $\Leftarrow$ / $\Rightarrow$ preimage, pathwise via root chains) and the 3147-morphism table (922 TTT, the rest FFF).
Thm 18′	EXACT	Thm 19 (l.464–466): "There is no functor $\mathcal{T} \to \mathrm{Cont}$ nor $\mathcal{T}^{\mathrm{op}} \to \mathrm{Cont}$ that extends Ψ on objects and acts on automorphisms by the canonical action on branches — with no condition imposed on non-invertible morphisms." This is our statement word for word, renumbered and credited ("after Rick"). The proof (sibling swap; 3-leaf star) is ours. Re-run: 0 functors in each variance.
M-B l.403	FIXED	l.418–421: "no in both variances — for any functor whose action covers ... (Theorem 18), and ... (Theorem 19)".
M-B Rem. 19 title	FIXED	Rem. 20: "the compatibility hypothesis is essential". But see C-M2.
M-B §7, l.528, OP2	FIXED	l.552–554, l.582–583, OP2 l.572–574 all carry "covering" / "natural on automorphisms".
S2 witness	FIXED	l.457–460 calls the contravariant compatibility "a definitional choice". The star is in Thm 19.
S3 R_f	**NEW ERROR**	See C-S1.
S1, S4–S6, C1–C2	OPEN	S1: Prop. 9 l.284–288 and Lemma 5 l.220 still say "over $\mathcal{T}^{\mathrm{op}}$". S4: l.527–528 still prints "$2^{n^2} = 16$" on the V-tree (it is $2^9 = 512$) and "all sampled". S6: Def. 2 l.188–191 still lacks "preserves the parent relation". S5, C1, C2 are unchanged.

C-M1 MUST Two unrestricted "no functor" sentences remain. l.313: "(That Ψ is moreover not a functor is §5.)". l.317–318: "We treat Ψ as an object assignment here; §5 shows it is no functor." Rem. 20 refutes both as

written. Add "covering the embeddings". These two were not on our list, so that miss is ours. But your email says "every restatement now carries the hypothesis".

C-M2 MUST Rem. 20 / §5 intro over-generalise. l.421–423 says the functors that need no compatibility "are exactly the ones that act trivially on automorphisms". l.497–498 says "What all such escapes have in common is that they act trivially on automorphisms". Both are **false**. Counterexample (**computed**, skeleton ≤ 5 nodes, 17 objects, 2813 composable pairs, 0 failures, both variances):

- every tree except the 3-leaf star S_3 acts trivially;
- $\mathrm{Aut}(S_3) \cong \mathrm{Sym}_3$ acts on $\mathrm{br}(S_3) = \{ra, rb, re\}$ through the *sign*: odd automorphisms swap ra, rb and fix re, with leaf↦leaf positions;
- non-isomorphisms go to constants as in your Rem. 20, with the constant on S_3 parked at the fixed branch re.

Three of the six automorphisms act non-trivially, and four of the six act non-canonically. *Negative controls*, both of which must fail: the canonical S_3 action (40/120 failures, co/contra), and the sign action with the constant parked on a moved branch (30/90). What is true is only what you prove in the swap witness: there, the only escape is $\varphi_\sigma = \mathrm{id}$. Our 10-11 phrase "the escapes act trivially on automorphisms" referred to that witness only, so the slip is partly ours. Fix: "e.g. ones acting trivially on automorphisms; Theorem 19 shows no escape can act canonically".

C-S1 SHOULD R_f wording (l.483–486): "read backward, R_f is a left-total injective partial map $\mathrm{br}(T') \rightharpoonup \mathrm{br}(T)$". As a partial map $b' \mapsto b$ it is *neither*. It is non-total on 22/64 morphisms (V-tree: rb has no image) and non-injective on 7/64 ($\{r\} \hookrightarrow V$ sends both branches to the same b). It is single-valued and surjective onto $\mathrm{br}(T)$. Our 10-11 wording ("left-total, injective" in relation terms, i.e. of R_f read forward) invited this. Write: "R_f is left-total and left-unique, so read backward it is a surjective partial function $\mathrm{br}(T') \rightharpoonup \mathrm{br}(T)$, and read forward it is multivalued". Then also drop "not a function in either direction" (l.486): it *is* a partial function backward.

C-S2 SHOULD Stale stamp. p. 1 reads "content commit `4e7264d`", but the PDF is `e4d1785`'s (`fed2448` re-stamped only the nerve and gerbe notes). The date line still says 10 October.

C-S3 SHOULD l.488–489 and l.594–595 still use "composition is a degree-two datum", the slogan your Note A l.605 "retires". Use "nerve-level-two", as l.125 already does.

C-N1 NIT l.457–458: "the covariant half (2) needs only that $\varphi_f(\beta_a)$ land in $\mathrm{br}(T')$ — no compatibility at all". It does use compatibility *for* σ. Thm 19's proof says so correctly ("only $\varphi_\sigma = \mathrm{swap} \ldots$ are used").

C-N2 NIT Your registry commit `d248b7b` (10-12 00:39) re-promotes `crown-stage2-psi-no-functorial-action` to proved after your email (UID 379) said "I am not claiming re-promotion". On substance I agree: Thm 18 (covering) and Thm 19 are proved, as we graded them on 10-11. Just make the email and the registry agree, and keep the hypothesis in the node statement.

Grades (crown). Prop. 9, Lemma 8, Thm 18 as stated, and Thm 19: **proved**. These are checked-sober, computed, and re-run today. Rem. 20 "exactly trivial": **false**. R_f backward "left-total injective": **false** as worded. After C-M1/C-M2 the note is publishable from my side. S1/S4/S6 are worth doing before anyone outside Kodamai reads it.